

Lab No 6:

Date:

Collective Communications and Error Handling in MPI

Objectives:

In this lab, student will be able to

1. Understand the different aggregate functions used in MPI
2. Learn how to Implement MPI programs using both point to point and collective communication routines
3. Learn and use the APIs for handling errors in MPI

I.Aggregation Functions

MPI provides two forms of aggregation

➤ Reduction

```
MPI_Reduce (SendAddress, RecvAddress, Count, Datatype, Op, Root, Comm);
```

Scan

Reduction: This routine reduces the partial values stored in **SendAddress** of each process into a final result and stores it in **RecvAddress** of the **Root** process. The reduction operator is specified by the **Op** field. Some of the reduction operator available in MPI are: MPI_SUM, MPI_MAX, MPI_MIN, MPI_PROD

Scan:

```
MPI_Scan (SendAddress, RecvAddress, Count, Datatype, Op, Comm);
```

This routine combines the partial values into N final results which it stores in the RecvAddress of the N processes. Note that root field is absent here. The scan operator is specified by the Op field. Some of the scan operator available in MPI are: MPI_SUM, MPI_MAX, MPI_MIN, MPI_PROD.

MPI_Barrier(Comm) :This routine synchronizes all processes in the communicator Comm. They wait until all N processes execute their respective MPI_Barrier.

Note: All collective communication routines except MPI_Barrier, employ a standard blocking mode of point-to-point communication.

Error Handling in MPI:

- An MPI communicator is more than just a group of process that belong to it. Amongst the items that the communicator hides inside is an error handler. The error handler is called every time an MPI error is detected within the communicator.
- The predefined default error handler, which is called `MPI_ERRORS_ARE_FATAL`, for a newly created communicator or for `MPI_COMM_WORLD` is to abort the whole parallel program as soon as any MPI error is detected. There is another predefined error handler, which is called `MPI_ERRORS_RETURN`.
- The default error handler can be replaced with this one by calling function `MPI_Errhandler_set`, for example:

```
MPI_Errhandler_set(MPI_COMM_WORLD, MPI_ERRORS_RETURN);
```

- The only error code that MPI standard itself defines is `MPI_SUCCESS`, i.e., no error. But the meaning of an error code can be extracted by calling function `MPI_Error_string`. On top of the above MPI standard defines the so called error classes. The error class for a given error code can be obtained by calling function `MPI_Error_class`.
- Error classes can be converted to comprehensible error messages by calling the same function that does it for error codes, i.e., `MPI_Error_string`. The reason for this is that error classes are implemented as a subset of error codes.

Solved Example:

Implement a MPI program using N processes to find $1! + 2! + \dots + N!$. Use collective communication routines.

```
#include <stdio.h>
#include "mpi.h"
int main(int argc, char* argv[])
{
    int rank, size, fact=1, factsum, i;
```

```

MPI_Init(&argc,&argv);
MPI_Comm_rank(MPI_COMM_WORLD,&rank);
MPI_Comm_size(MPI_COMM_WORLD,&size);

for(i=1; i<=rank+1; i++)
    fact = fact * i;

MPI_Reduce (&fact,&factsum, 1, MPI_INT, MPI_SUM, 0, MPI_COMM_WORLD);

if(rank==0)
    printf(" Sum of all the factorial=%d",factsum);

MPI_Finalize();
exit(0);
}

```

Lab Exercises:

- 1) Implement a MPI program using N processes to find $1! + 2! + \dots + N!$. Use scan. Also, handle different errors using error handling routines.
- 2) Implement a MPI program to read a 3 X 3 matrix. Enter an element to be searched in the root process. Find the number of occurrences of this element in the matrix using three processes.
- 3) Implement a MPI program to read 4 X 4 matrix and display the following output using four processes.

I/p matrix: 1 2 3 4
 1 2 3 1
 1 1 1 1
 2 1 2 1

O/p matrix: 1 2 3 4
 2 4 6 5
 3 5 7 6
 5 6 9 7

- 4) Implement parallel odd even transposition sort

Additional Exercises:

1. Implement a MPI program to read a word of length N. Using N processes including the root get output word with the pattern as shown in example. Display the resultant output word in the root.

Example: Input : PCAP

Output : PCCAAAPPPP

2. Implement a MPI program to read matrix A of size 5 X 5. It produces a resultant matrix B of size 5 X 5. It sets all the principal diagonal elements of B matrix with 0. It replaces each row elements in the B matrix in the following manner: If the element is below the principal diagonal it replaces it with the maximum value of the

column in the A matrix having the same row number of B. If the element is above the principal diagonal it replaces it with the minimum value of the column in the A matrix having the same row number of B. Produce the B Matrix using 5 processes. Use only Collective communication routines except broadcast routine.

Example:

A				
1	2	3	4	5
5	4	3	2	1
10	3	13	14	15
11	22	11	33	44
1	12	5	4	6

B				
0	1	1	1	1
22	0	2	2	2
13	13	0	3	3
33	33	33	0	2